Skill Level 1

Maven User Manual — All Personnel

No technical background required • Prerequisite: None • Open PDF →

▸ BLUF

Skill Level 1 (SL 1) teaches you how to log in, navigate MSS, read dashboards, use Gaia and command-level applications, submit data through forms, interact with AI tools embedded in Workshop, and stay within authorized boundaries. Required for all USAREUR-AF personnel before operating MSS. **Request your account before your first class — submit at mss.data.mil or through your data steward. Provisioning generally completes within 24 hours.**

◆ PUBLICATION

TM-10 (SL 1) — Maven User Manual — Full technical manual. Open PDF →

⚑ CAUTION — REQUEST YOUR ACCOUNT FIRST

You cannot log in without a provisioned account. **Do not wait until the first day of class.** Request your account at mss.data.mil or through your unit data steward — provisioning generally completes within 24 hours. If access is not active after 24 hours, contact your data steward. Steps are in Section 1 below.

1. GETTING ACCESS — DO THIS FIRST

1. Find your **unit data steward** (data stewards may be embedded in your unit, assigned at battalion, brigade, or division level, or positioned within a directorate at the ASCC — ask your chain of command if you're unsure who to contact).
2. Ask them to submit an MSS account request with your name, unit, MOS, and required access level.
3. MSS admin team provisions your account and assigns **markings** (markings = the data categories and classification levels you're authorized to see).
4. You receive notification when account is active — typically 3–5 business days.
5. Receive the MSS portal URL from your unit data steward. This is your login link.

• NOTE — DATA STEWARD

Your **data steward** manages MSS access and data quality for your organization. They may be embedded in your unit, assigned at a higher echelon, or positioned within a directorate at the

ASCC. They are your first point of contact for account requests, access problems, and data errors. If you don't know who they are, ask your chain of command.

2. WHAT IS MSS?

MSS is the mission command information system (MCIS) program of record, directed by the USAREUR-AF CG to enable rapid and accurate decision-making. It is a secure, web-based platform where your unit's data lives and can be analyzed and acted upon. Think of it as a shared operations center for data: information from logistics, personnel, readiness, and other systems is collected, organized, and made accessible through applications your unit uses every day.

MSS is built on the Palantir Foundry platform, authorized for Army use under the Maven Smart System program.

• NOTE — WHAT IS FOUNDRY?

Palantir Foundry is a commercial data platform that Army headquarters selected to run MSS. You do not need to know how it works — just how to use it. The word "Foundry" may appear in help documentation and system menus; it refers to the same platform as MSS.

MSS DOES

Stores data from Army systems in a single, organized location; makes data visible through applications and dashboards; enables units to update records, report status, and track readiness; provides analysis tools for authorized personnel; supports AI-assisted analysis through AIP tools embedded in applications; includes **Gaia** — a map-based geospatial application for situational awareness and operational overlays; and provides command-level applications (e.g., CUB/CUA in USAREUR-AF) for operational briefing and C2.

MSS IS NOT

Not a replacement for official systems of record (DCPDS, GCSS-A, MEDPROS, etc.); not classified by default — classification depends on data markings; not a public system — access is controlled and audited.

USAREUR-AF MISSION AREAS

- **Personnel Readiness** — Soldier readiness status

- **Logistics** — equipment availability & maintenance
- **Operational Reporting** — SITREPs and updates
- **Planning** — orders, unit positions, task org
- **C2** — unit status across the AOR

3. SECURITY RESPONSIBILITIES

⚠ **WARNING**

Unauthorized access to, disclosure of, or modification of data in MSS may constitute a violation of 18 U.S.C. § 1030 (Computer Fraud and Abuse Act) and applicable Army regulations. Violations may result in disciplinary action, loss of access, and criminal prosecution.

1. Use only your own credentials. Do not share your CAC, PIN, or access tokens.
2. Access only data you are authorized to view.
3. Report misrouted data immediately. If you see data at a higher classification than your clearance — STOP and report it.
4. Do not export data without authorization. Exports are logged.
5. Log out when done. Do not leave an MSS session unattended on an unlocked workstation.
6. Report security incidents immediately to your supervisor and unit security officer.

4. TASKS

TASK: ACCESS THE MAVEN SMART SYSTEM

CONDITIONS	STANDARDS	EQUIPMENT
Provisioned MSS account, CAC reader, approved workstation and browser	Successfully authenticate with CAC and reach the MSS home screen	CAC, CAC reader, workstation, MSS portal URL (from unit data steward)

1. Insert your CAC into the CAC reader.
2. Open an approved web browser (Chrome or Firefox recommended).
3. Navigate to the MSS portal URL provided by your unit data steward.
4. When prompted, select your **authentication certificate** (not email certificate).
5. Enter your CAC PIN when prompted.

6. MSS home screen loads — you are now logged in.

> ⚑ **CAUTION**
>
> Do not save your PIN in the browser. Do not allow the browser to remember your login. MSS sessions may contain sensitive information.

TASK: NAVIGATE THE MSS HOME SCREEN

CONDITIONS	STANDARDS
Logged into MSS	Identify all major navigation elements; locate search, notifications, and profile

ELEMENT	LOCATION	PURPOSE
Search bar	Top center	Find datasets, applications, and projects
Notification bell	Top right	System alerts, workflow updates
User profile icon	Top right	Account settings, markings (your authorized data categories), logout
Compass (file explorer)	Left sidebar	Browse all MSS resources; pin resources to top of Files page for quick access
Home button (logo)	Top left	Return to home screen from anywhere
Pinned items	Home main area	Shortcuts to frequently used resources
Recent activity	Home main area	Recently visited datasets and apps
Notepad	Left sidebar / search	Draft documents with AIP-assisted editing — use custom prompts and recently-used functions to accelerate writing

5. Q1 2026 PLATFORM UPDATES

• WHAT'S NEW — Q1 2026

Palantir shipped several updates that affect SL 1 operator tasks. Instructors should incorporate these into practical exercises.

WORKSHOP

- **AIP Analyst sessions persist** when you switch tabs or change sections — no more losing your chat context
- **Custom background colors** on Workshop sections and pages for better visual organization
- **Markdown editing** now available in text input widgets
- **Microsoft Word export** — export Workshop content directly to .docx (in addition to CSV)
- **Usage metrics** — track how Workshop applications are being used across your organization
- **Custom widgets on mobile** — Workshop mobile now supports custom widget rendering

NOTEPAD

- **Notepad** is a document editor integrated into MSS for drafting reports, SOPs, and notes
- **AIP-assisted editing** — enter custom prompts to have AIP help draft, revise, or restructure your text
- **Recently-used functions** appear in the AIP Assist menu for quick reuse
- **AI FDE integration** — Notepad connects to the AI forward deployed engineer for deeper analysis

GAIA & COMPASS

- **Gaia default styles** — interfaces with geometry properties now have configurable default map styles in Ontology Manager
- **Gaia → Workflow Lineage** — new shortcut to jump from a Gaia map directly to Workflow Lineage for tracing data sources
- **Compass pinning** — pin resources to the top of the Files page for quick access to frequently-used items

- **Project selection** now available during installation configuration

6. REPORTING PROBLEMS

PROBLEM TYPE	WHO TO CONTACT
Cannot log in	MSS Help Desk
Cannot access a project	Unit data steward
Data appears incorrect	Unit data steward (do not correct it yourself)
System error or crash	MSS Help Desk (provide error code and screenshot)
Security incident	Supervisor and unit security officer — IMMEDIATELY
Application not working	MSS Help Desk

SCHED UPCOMING TRAINING — SL 1

• ENROLLMENT

Contact the listed POC to reserve a seat. Bring your CAC and ensure your MSS account request is submitted at least 5 business days before the course start date. Virtual sessions require MS Teams access and a headset.

DATES	LOCATION	FORMAT	POC	SEATS	STATUS
14 APR 2026	Wiesbaden, Clay Kaserne, Bldg 3312, Rm 104	In-Person	SSG Johnson	20 / 20	**OPEN**
05 MAY 2026	Grafenwöhr, Bldg 244, Conf Rm B	In-Person	SFC Davis	8 / 20	**8 SEATS REMAINING**
18 JUN 2026	Stuttgart, Kelley Bks, Bldg 3357	In-Person	SSG Martinez	20 / 20	**OPEN**

DATES	LOCATION	FORMAT	POC	SEATS	STATUS
09 JUL 2026	Virtual (MS Teams)	Virtual	SSG Johnson	30 / 30	**OPEN**

Duration: 1 day (8 hours). Course runs 0800–1700. All dates subject to change — confirm with POC 5 days prior.

NEXT LEVEL — AFTER SL 1

TM-20 (SL 2) — No-Code Builder Manual

Learn to ingest data, build Workshop applications, create Object Types, and manage projects — all without coding. Required prereq for all builder roles.

Continue to SL 2 →

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →** For task-level procedures, use the **Task Index →**

SL 2

No-Code Builder Manual — All Staff

Prerequisite: SL 1 • No coding required •
Open PDF →

▶ BLUF

SL 2 teaches you how to ingest data, build Workshop applications, create Object Types and basic Actions, and manage projects — all using the graphical user interface. No coding required. Prerequisite: SL 1 complete.

◆ PUBLICATION

TM-20 (SL 2) — No-Code Builder Manual — Full technical manual. Open PDF →

COMPETENCIES UPON COMPLETION

PROJECT MANAGEMENT (UI)

- Use **Solution Designer** to visually map data flows and application architecture before building
- Build project roadmaps using forward and backward planning from the commander's requirement
- Create and organize Foundry projects via the UI
- Set up folder structure: raw / staging / curated layers
- Manage project access and permissions via UI
- Follow USAREUR-AF naming conventions and builder standards

DATA INGEST (NO CODE)

- Ingest data using Pipeline Builder — visual, no code
- Configure connectors and file sources via UI
- Schedule pipeline runs via UI
- Understand raw / staging / curated dataset layers

ONTOLOGY (UI)

- Create Object Types and set primary keys via Ontology Manager UI
- Define **Interfaces** and apply them to Object Types for consistent property contracts

WORKSHOP APPLICATIONS

- Build and publish Workshop applications with dashboards, forms, and filters
- Select and configure appropriate widgets for each use case

- Configure **Object Views** to control how properties display to operators
- Create Link Types between objects via UI
- Understand Action types (write-back, form, webhook, conditional) and configure basic Actions
- Validate Object data in **Object Explorer** before building apps

- Apply access controls and publish to users

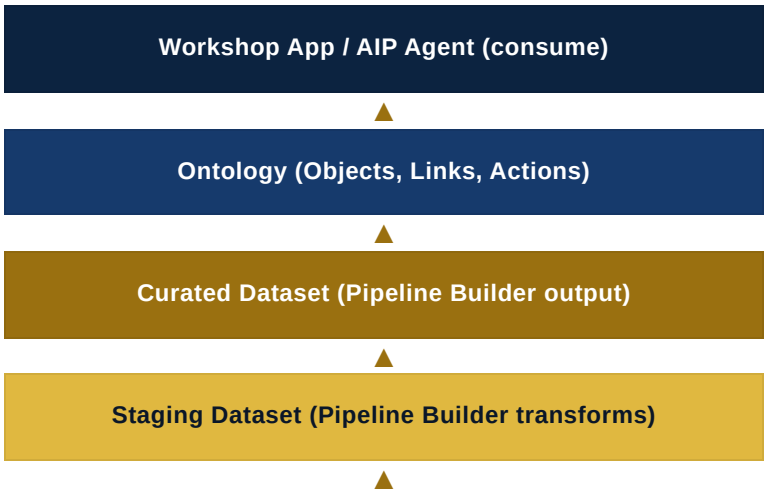
BRANCHING & GOVERNANCE

- Use Foundry branching to build and promote via UI
- Distinguish development from production environments
- Apply USAREUR-AF builder standards

*Branching = making a test copy of your work before going live. You build in the **dev branch** (your sandbox), test it, then publish to **production** (what users see).*

THE FOUNDRY DATA STACK

Data flows through layers. As a SL 2 builder, you work in the middle layers using visual tools. Never modify raw data — report data errors to your data steward instead.



Raw Dataset — READ ONLY, never modify

WORKSHOP WIDGET SELECTION

YOU NEED TO…	USE THIS WIDGET
Display many objects or records	Object Table
Show details for one selected object	Object Detail
Let users filter the data they see	Filter Panel / Dropdown
Show a chart (bar, line, pie)	Chart Widget
Show geographic data on a map	Map Widget
Let users write or update data	Button + Action or Action Form
Show a single key metric prominently	Metric Tile
Navigate between app sections	Navigation / Tab Widget

ONTOLOGY SETUP ORDER (UI STEPS)

1. Confirm curated dataset exists and is populated (Pipeline Builder pipeline passing)
2. Open **Ontology Manager** in the left sidebar
3. Create Object Type → set primary key property → map properties from curated dataset
4. Create Link Types between related Object Types (if needed)
5. Publish ontology branch and test in **Object Explorer**
6. Build Workshop app only after Object Explorer confirms objects are visible

NAMING CONVENTIONS

OBJECT TYPE	CONVENTION	EXAMPLE
Datasets (path)	`/Project/AOR/source/raw|staging|curated`	`/USAREUR/EUR/personnel/c`
Object Types	`PascalCase`	`UnitStatus`

OBJECT TYPE	CONVENTION	EXAMPLE
Properties (API name)	`camelCase`	`unitName`
Properties (display name)	Title Case	`Unit Name`
Link Types	`PascalCase` verb form	`HasEquipment`
Workshop app names	Unit + function + version	`EUR-Personnel-Readiness-`

⚑ CAUTION — SHARED RESOURCES

Changes to shared datasets and Object Types affect all downstream applications and users. Before modifying a shared resource, coordinate with your data steward.

Q1 2026 PLATFORM UPDATES

• WHAT'S NEW — Q1 2026

Palantir shipped several updates that affect SL 2 builder tasks. Instructors should incorporate these into practical exercises.

OBJECT VIEWS — NOW GA

- **Core Object Views** are now generally available (Feb 2026) — standard, configurable views for browsing and exploring object data
- **Foundry Branching support** — develop and test Object Views safely in branches before promoting to production
- **Updated Object Explorer** with redesigned landing page for improved ontology discoverability

PIPELINE BUILDER

- **Incremental execution enforcement** — configure Pipeline Builder to require incremental execution for efficiency
- **Preview behavior controls** — enforce manual confirmation before running previews to prevent accidental full runs
- **LLM data generation** — generate notional/test data with LLMs directly in Pipeline Builder
- **Approximate nearest neighbor join** for vector similarity and geospatial matching

- **File/file set output** — output files and file sets directly from Pipeline Builder

BRANCHING & WORKFLOW

- **Role-based branch security** — assign users roles to control who can manage and merge branches
- **Upgraded branch security** now enabled for all users by default
- **Ontology SQL** in Quiver for querying object sets directly
- **Quiver time series** — dedicated workspace for time series analysis

SCHED UPCOMING TRAINING — SL 2

• ENROLLMENT

SL 1 must be complete before attending SL 2. Contact the listed POC to reserve a seat. Bring completion certificate from SL 1 on day one.

DATES	LOCATION	FORMAT	POC	SEATS	STATUS
21–25 APR 2026	Wiesbaden, Clay Kaserne, Bldg 3312, Rm 104	In-Person	SFC Chen	7 / 15	**7 SEATS REMAINING**
11–15 MAY 2026	Grafenwöhr, Bldg 244, Conf Rm B	In-Person	SSG Williams	3 / 15	**3 SEATS REMAINING**
22–26 JUN 2026	Stuttgart, Kelley Bks, Bldg 3357	In-Person	SFC Chen	15 / 15	**OPEN**
13–17 JUL 2026	Virtual (MS Teams)	Virtual	SSG Williams	20 / 20	**OPEN**

Duration: 5 days (40 hours). Course runs 0800–1700 each day. Prerequisite: SL 1 complete. All dates subject to change — confirm with POC 5 days prior.

NEXT LEVEL — AFTER SL 2

TM-30 (SL 3) — Advanced Builder Manual

For data-adjacent specialists: complex application design, advanced pipelines, Ontology architecture, AIP Logic, and C2DAO governance. Required prereq for all SL 4 tracks.

Continue to SL 3 →

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →** For task-level procedures, use the **Task Index →**

FBC

Foundry Bootcamp — Applied Build Event

Prerequisite: SL 2 + command-validated project • Quarterly • Outside TM chain • No TM credit granted

▶ BLUF

The Foundry Bootcamp is a quarterly 5-day supervised build event. You bring a validated operational problem; you build a solution; SMEs are available for consultation. Minimal instruction — this is not a course. You leave with a functional product and a handoff package. FBC does not replace SL 3 and does not grant credit toward SL 4 enrollment.

◆ PUBLICATION

FBC — Foundry Bootcamp Participant Guide — Full reference document. **Open PDF →**

⚠ NOTE

FBC is **outside the SL 1 through SL 5 training chain.** Completion does not count as SL 3 or any other TM credit. If you need structured platform instruction, enroll in SL 3. FBC is for builders with SL 2 skills and a real problem to solve.

WHO ATTENDS

REQUIRED

- **SL 2 Go on file** — hard requirement, no exceptions
- **Command-approved Project Brief** — submitted to C2DAO ≥14 days before Day 1
- Supervisor signature on enrollment request

PROJECT REQUIREMENTS

- Specific output: named dashboard, pipeline, Ontology type, or Quiver/Contour product
- Named consumer — a real person or role who will use the product
- All data sources accessible *before* Day 1
- No code required — Python / TypeScript / OSDK = SL 4 track, not FBC
- 5-day feasibility: functional prototype reachable within sprint

SPRINT WEEK STRUCTURE

DAY	ACTIVITY
Day 1	In-brief: scope review, environment check, kickoff (0800–0900). Build (0900–1700).
Days 2–4	Daily standup (0800, 15 min). Build (0815–1700). SME available throughout.
Day 5	Product demo / peer review (0800–1000). Go/No-Go determination (1000–1200). Out-brief and handoff (1300–1500).

GO STANDARD

STANDARD	CRITERION
Functional product	The product does what your Project Brief says it will do — your named consumer can use it
Documentation	Naming conventions followed; product description explains purpose and data sources
Handoff package	Complete by end of Day 5 — product description, data sources, limitations, maintenance guidance, promotion status, POC
Governance	Product in a branch; promotion plan documented or production promotion initiated

ENROLLMENT

TIMELINE

- **T-21 days:** Enrollment request submitted
- **T-14 days:** Project Brief approved by C2DAO
- **T-10 days:** Sprint workspace provisioned
- **T-5 days:** Candidate confirms access
- **Day 1:** Sprint begins

CADENCE

- 4 sprint events per fiscal year (quarterly)
- 4–16 participants per sprint
- 1 SME per ≤8 participants
- Annual schedule published each October

◆ **DOCUMENTS**

Participant Guide: **FBC_GUIDE.pdf** • Coordinator Package: **FBC_SPRINT_PACKAGE.pdf** • SOP: **FOUNDRY_BOOTCAMP_SOP.pdf** • Environment Setup: **FBC_ENVIRONMENT_SETUP.pdf** • Project Brief form: **CAD Appendix D**

EXEC

Senior Leader Executive Course

Audience: O-5+ / E-9+ • 1 day • No prerequisites • Terminal — outside the TM pipeline

▶ BLUF

EXEC gives battalion commanders, command sergeants major, and equivalent senior leaders the operational understanding of MSS required to lead formations that depend on data-driven decision-making. You will not build anything. You will learn what the platform produces, how data products affect your formation, and how to direct your staff's use of data as a command function. EXEC replaces SL 1 for O-5 / E-9+ personnel. It is terminal — no progression to SL 2 or beyond.

◆ PUBLICATION

TM-EXEC — Senior Leader Executive Course — Full reference document. **Open PDF →**

⚠ NOTE

EXEC is **outside the SL 1 through SL 5 training chain.** It does NOT grant SL 1 credit. If a senior leader wants hands-on platform qualification, they should enroll in SL 1 and proceed through the standard pipeline.

WHAT THIS COURSE COVERS

YOU WILL LEARN

What MSS does for your formation and why it matters; how to evaluate data products — operationally, not technically; how to guide your formation's data posture through resourcing, prioritization, and governance; what questions to ask about data freshness, source integrity, and product quality; and the training pipeline that qualifies your data workforce (SL 1 through SL 5).

THIS COURSE IS NOT

Not a platform navigation course — you will see MSS, not operate it; not a data literacy primer — you already understand why data matters; not a substitute for SL 1 in the standard pipeline; not a qualification to build, modify, or administer anything on the platform.

DAILY SCHEDULE

TIME	BLOCK	CONTENT
0800–0830	1	Course introduction; senior leader role in the data environment
0830–0930	2	Why MSS exists — strategic context, CG guidance (Ch 1)
0930–1030	3	The platform and what it produces — five-layer architecture, data product types, live walkthrough (Ch 2)
1030–1045	—	Break
1045–1200	4	How data products impact your formation — data as command function, failure patterns, Commander's Data PIRs (Ch 3)
1200–1300	—	Lunch
1300–1345	5	The training pipeline — SL 1 through SL 5, FBC, resourcing decisions (Ch 4)
1345–1430	6	Governance — the governance chain, VAUTI framework, red flags (Ch 5)
1430–1445	—	Break
1445–1530	7	How data projects work — agile overview, roadmap vs POAM (Ch 6)
1530–1615	8	Working with data professionals — engagement practices, terminology (Ch 7)
1615–1700	9	Asking the right questions — diagnostic questions for products, workforce, and AI (Ch 8–9)

DOCUMENTS

◆ TM-EXEC PUBLICATIONS

Course Manual: **TM_EXEC_SENIOR_LEADER.pdf** • Concepts Guide: **CONCEPTS_GUIDE_TM_EXEC.pdf** • Syllabus: **SYLLABUS_TM_EXEC.pdf**

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →**

SL 3

Advanced Builder Manual — Data-Adjacent Specialists

Prerequisite: SL 1 + SL 2 • 17/25-series • G2 • Data analysts • **Open PDF →**

▶ BLUF

SL 3 is for personnel who design and own MSS solutions. This level covers complex application design (including Kairos timelines and Target Workbench), advanced Pipeline Builder, Ontology architecture, AIP Logic configuration, basic agent building in Agent Studio, Automations, Machinery, data governance, and C2DAO standards. All work is done via the UI; advanced coding and agent development are escalated to SL 4 developers.

◆ PUBLICATION

TM-30 (SL 3) — Advanced Builder Manual — Full technical manual. **Open PDF →**

⚑ CAUTION

Modifications to shared datasets or Object Types at SL 3 level affect all downstream applications and users across the formation, including coalition partners. Coordinate with your unit data steward and the USAREUR-AF C2DAO before publishing any changes to production resources.

• NOTE — IS THIS YOUR LEVEL?

SL 3 covers advanced no-code building — application design, pipeline architecture, governance. If your role requires **coding, ML, or ORSA**, SL 3 is a prerequisite to a specialist track.
View Specialist Tracks (SL 4/5) →

COMPETENCIES UPON COMPLETION

ADVANCED WORKSHOP DESIGN

- Design multi-page Workshop applications with conditional logic and variable passing
- Build dynamic layouts: show/hide panels based on user selections
- Design navigation flows and inter-page parameter handoff

ADVANCED PIPELINE BUILDER

- Build multi-source join pipelines with complex aggregations (visual)
- Design scheduled and triggered pipeline runs
- Review and interpret data lineage graphs
- Escalate to SL 4 when code transforms are required

- Publish and manage application versions

ONTOLOGY ARCHITECTURE

- Design Object Type and Link Type models via Ontology Manager UI
- Architecture thinking: model for downstream app requirements, not just source data
- Design Action workflows with validation and approval logic via UI
- Coordinate ontology changes with all downstream application owners

ADVANCED ANALYSIS

- Conduct advanced Contour analysis: complex aggregations, pivots, calculated columns, saved views
- Build advanced Quiver dashboards with multi-object analysis and linked views
- Create reusable analysis templates for unit use

AIP & AGENT STUDIO

- Configure AIP Logic workflow parameters and natural language query on Object Types
- Build **basic AIP agents** in Agent Studio — connect Object Types, write output instructions, test and validate responses
- Orient to AIP Analyst and AI FDE — awareness of the full AIP toolset
- Advanced agent development (custom tools, action logic, production deployment) is SL 4H scope

KAIROS & TARGET WORKBENCH

- Configure **Kairos** timeline widgets — Ontology-driven, real-time planning visualization (distinct from static Gantt charts)
- Orient to **Target Workbench** — targeting workflow integration; operational use is SL 4A/SL 4B scope
- Ensure Object Types are structured to support both tools

AUTOMATIONS & MACHINERY

- Configure **Automations** on Object Types — schedule or condition-triggered property updates, pipeline runs, and notifications
- Model basic business processes in **Machinery** — multi-step workflows with roles, transitions, and object linkage

GOVERNANCE & PRODUCTION

- Manage branching and production promotion via UI
- Apply USAREUR-AF C2DAO governance standards and naming conventions
- Manage governance workflows with data stewards
- Ensure coalition-facing products have C2DAO coordination and NAFv4

- Monitor automation execution history and active process instances

compliance review

SL 3 vs SL 4 — WHAT YOU OWN VS WHAT YOU ESCALATE

YOU OWN AT SL 3 (UI)	ESCALATE TO SL 4 WHEN…
Application design and UX	Custom Python/PySpark transforms needed
Ontology model design (via UI)	Functions on Objects (TypeScript) required
Advanced Pipeline Builder (visual)	Incremental watermark or code logic needed
AIP Logic configuration + basic agent building	Custom tool development, agentic action logic, or production agent deployment needed
Governance coordination	External application (OSDK) needed
Production promotion via UI	CI/CD pipeline automation needed

C2DAO GOVERNANCE GATES — HARD STOPS

REQUIREMENT	SL 3 ACTION	HARD GATE?
New shared Object Type or dataset	Coordinate with C2DAO before publishing to production	Yes
Coalition / MPE-facing data product	C2DAO coordination + NAFv4 compliance review	Yes — do not skip
Schema change to existing shared resource	Notify all downstream owners; coordinate with steward	Yes
New AIP Logic workflow on operational data	Authorization review before deployment	Yes
Access permission changes	Submit through formal request to unit data steward	Yes

SCHED UPCOMING TRAINING — SL 3

• ENROLLMENT

SL 1 and SL 2 must be complete before attending SL 3. Class size is limited. Contact POC early — seats fill quickly. Bring SL 1 and SL 2 completion certificates on day one.

DATES	LOCATION	FORMAT	POC	SEATS	STATUS
28 APR – 02 MAY 2026	Wiesbaden, Clay Kaserne, Bldg 3312, Rm 104	In-Person	CW3 Thompson	4 / 10	**4 SEATS REMAINING**
15–19 JUN 2026	Virtual (MS Teams)	Virtual	CW2 Rodriguez	15 / 15	**OPEN**
17–21 AUG 2026	Wiesbaden, Clay Kaserne, Bldg 3312, Rm 104	In-Person	CW3 Thompson	10 / 10	**OPEN**

Duration: 5 days (40 hours). Course runs 0800–1700 each day. Prerequisites: SL 1 and SL 2 complete. All dates subject to change — confirm with POC 5 days prior.

Q1 2026 PLATFORM UPDATES

• WHAT'S NEW — Q1 2026

These Foundry platform updates affect SL 3 advanced builder tasks. Instructors should integrate into labs and exercises.

AIP LOGIC & AUTOMATIONS

- **AIP Logic branch selection** for evals — run evals against specific branches to test changes before promotion
- **Autopilot** (new, Mar 2026) — visualize, monitor, and debug automation workflows in a dedicated UI
- **Interface parameters** now supported in Automate actions
- **Function effects** supported in Automate — actions can now trigger

WORKFLOW LINEAGE

- **Presentation mode** — create and present lineage visualizations with frames
- **Multi-ontology support** — view lineage across multiple ontologies in one graph
- **Log search** — search logs directly from Workflow Lineage nodes
- **Monitoring status colors** — color lineage nodes by monitoring status for

function side effects

- **Streaming time series conditions** — create conditions on streaming time series data in Automate

quick health assessment

- **Expanded access** — Workflow Lineage is now accessible from more locations across the platform (Gaia, Quiver, Notepad, Automate)

BRANCHING & GOVERNANCE

- **Role-based branch security** — assign users specific roles (manage, merge) for fine-grained branch access control
- **Upgraded branch security** enabled for all users — all new branches use the enhanced security model
- **Object Views + Branching** — develop and test Object Views safely in branches before production
- **Materializations** now supported for object types with row-level policies

NEXT LEVEL — AFTER SL 3

Specialist Tracks — SL 4 & SL 5 Series

Fourteen tracks: six Warfighting Function (WFF) tracks covering Intelligence, Fires, Movement & Maneuver, Sustainment, Protection, and Mission Command — plus eight specialist tracks for ORSA, AI Engineer, ML Engineer, Program Manager, Knowledge Manager, Software Engineer, UI/UX Designer, and Platform Engineer. Select your track based on MOS/role.

Access Specialist Tracks →

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →** For task-level procedures, use the **Task Index →**

SPECIALIST TRACKS

SL 4 & SL 5 Series — WFF & Technical Tracks

WFF tracks (A–F): SL 3 req. • Technical tracks (G–O): SL 3 req.

▶ BLUF

SL 4 has two track types: **Warfighting Function tracks (SL 4A–F)** for WFF-assigned roles (Intelligence, Fires, M&M, Sustainment, Protection, Mission Command) — prerequisite SL 3, no coding required — and **Technical Specialist tracks (SL 4G–O)** for personnel who build and engineer MSS solutions (ORSA, AI Eng, MLE, PM, KM, SWE, UX Designer, Platform Eng) — prerequisite SL 3. Advanced versions (SL 5G–O) are available after completing the corresponding SL 4 technical track.

TRACK SELECTION BY MOS / ROLE

ROLE / MOS	RECOMMENDED TRACK	ADVANCED
WARFIGHTING FUNCTION TRACKS (SL 4A–F)		
G2/S2 — MI units, ISR analysts	**SL 4A (Intelligence)**	—
FA officers/NCOs — Fire support	**SL 4B (Fires)**	—
Maneuver units — G3/S3 data roles	**SL 4C (Movement & Maneuver)**	—
G4/S4 — Logistics, GCSS-A	**SL 4D (Sustainment)**	—
Air defense, CBRN, force protection	**SL 4E (Protection)**	—
G6/S6 — C2 systems, networks	**SL 4F (Mission Command)**	—
TECHNICAL SPECIALIST TRACKS (SL 4G–O)		
FA49 — Operations Research Analyst	**SL 4G (ORSA)**	SL 5G
G2/S2 quantitative analyst	**SL 4G (ORSA)** or SL 4K (KM)	SL 5G / SL 5K
17A/17C — Cyber officer/NCO	**SL 4L (SWE)** or SL 4H (AI Eng)	SL 5L / SL 5H
25D — IT specialist	**SL 4L (SWE)**	SL 5L
AI/ML engineer (GS/contractor)	**SL 4H (AI Eng)** or SL 4M (MLE)	SL 5H / SL 5M

ROLE / MOS	RECOMMENDED TRACK	ADVANCED
Data scientist (GS/contractor)	**SL 4G (ORSA)** or SL 4M (MLE)	SL 5G / SL 5M
G8/S8 — Resource manager	**SL 4J (PM)**	SL 5J
Program Manager (PM / GS)	**SL 4J (PM)**	SL 5J
KMO / Knowledge Officer / 37F	**SL 4K (KM)**	SL 5K
Civil Affairs	**SL 4J (PM)** or SL 4K (KM)	SL 5J / SL 5K
UI/UX designer (GS/contractor)	**SL 4N (UX Designer)**	SL 5N
Platform engineer / DevOps / SysAdmin	**SL 4O (Platform Eng)**	SL 5O

SL 4 SERIES — WARFIGHTING FUNCTION TRACKS

TM-40A (SL 4A) — Intelligence

SL 3 REQ.

Intelligence Warfighting Function

G2/S2 • MI units • ISR analysts

PREREQ: SL 3 • OPEN PDF →

TM-40B (SL 4B) — Fires

SL 3 REQ.

Fires Warfighting Function

FA officers/NCOs • Fire support coordinators

PREREQ: SL 3 • OPEN PDF →

TM-40C (SL 4C) — Movement & Maneuver

SL 3 REQ.

Movement & Maneuver WFF

Maneuver units • G3/S3 data roles

PREREQ: SL 3 • OPEN PDF →

TM-40D (SL 4D) — Sustainment

SL 3 REQ.

Sustainment Warfighting Function

Logistics • G4/S4 • GCSS-A users

PREREQ: SL 3 • OPEN PDF →

TM-40E (SL 4E) — Protection

SL 3 REQ.

Protection Warfighting Function

Air defense • CBRN • Engineer • Force protection

PREREQ: SL 3 • OPEN PDF →

TM-40F (SL 4F) — Mission Command

SL 3 REQ.

Mission Command Warfighting Function

G6/S6 • C2 systems • Network managers

PREREQ: SL 3 • OPEN PDF →

SL 4 SERIES — TECHNICAL SPECIALIST TRACKS (Level 1)

TM-40G (SL 4G) — ORSA

SL 3 REQ.

Operations Research & Systems Analysis

FA49 • G2/S2 quant analysts • Wargame analysts

PREREQ: SL 3 • ADVANCED: SL 5G • OPEN PDF →

TM-40H (SL 4H) — AI Engineer

SL 3 REQ.

AIP Logic, Agent Studio & LLM Integration

AI/ML specialists • 17A/17C

PREREQ: SL 3 • ADVANCED: SL 5H • OPEN PDF →

TM-40M (SL 4M) — ML Engineer

SL 3 REQ.

Model Development, Validation & Deployment

ML engineers • Data scientists (GS/contractor)

PREREQ: SL 3 • ADVANCED: SL 5M • OPEN PDF →

TM-40J (SL 4J) — Program Manager

SL 3 REQ.

Pipeline Mgmt, Milestones & Portfolio Health

G8/S8 • PMs • Civil Affairs • 4 days

PREREQ: SL 3 • ADVANCED: SL 5J • OPEN PDF →

TM-40K (SL 4K) — Knowledge Manager

SL 3 REQ.

Forms, Lessons Learned & Institutional Memory

KMOs • 37F • Civil Affairs • 4 days

PREREQ: SL 3 • ADVANCED: SL 5K • OPEN PDF →

TM-40L (SL 4L) — Software Engineer

SL 3 REQ.

Python/TypeScript, OSDK & Code Transforms

17A/17C • 25D • GS/contractor SWEs

PREREQ: SL 3 • ADVANCED: SL 5L • OPEN PDF →

TM-40N (SL 4N) — UI/UX Designer

SL 3 REQ.

Soldier Centered Design, Workshop & Slate UI

UI/UX designers • Human factors • GS/contractor designers

PREREQ: SL 3 • ADVANCED: SL 5N • OPEN PDF →

TM-40O (SL 4O) — Platform Engineer

SL 3 REQ.

Kubernetes, CI/CD, DevSecOps & Infrastructure as Code

Platform engineers • DevOps • SysAdmins • 25D

PREREQ: SL 3 • ADVANCED: SL 5O • OPEN PDF →

SL 5 SERIES — ADVANCED TRACKS (Level 2)

◆ NOTE

SL 5 tracks are advanced-level continuations requiring the corresponding SL 4 track as a prerequisite. They cover expert-level techniques, production architecture, and operational integration at scale.

Open SL 5 Series Full Reference →

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →** For task-level procedures, use the **Task Index →**

SL 4

Technical Specialist Tracks — Developer Manuals

Prerequisite: SL 3 • Eight tracks by role • Advanced versions at SL 5

▶ BLUF

The SL 4 Technical Specialist tracks (SL 4G–O) cover developer-level capabilities requiring coding, advanced tooling, or specialized technical expertise — prerequisite SL 3. Warfighting Function tracks (SL 4A–F) are no-code, operationally focused training covering Intelligence, Fires, M&M, Sustainment, Protection, and Mission Command MSS integration — prerequisite SL 3. Select your track using the MOS/Role table on the HOME tab.

TM-40G (SL 4G) — ORSA Track

SL 3 REQUIRED

Operations Research & Systems Analysis

FA49 • G2/S2 quantitative analysts • Wargame analysts

- Configure Code Workspaces (Python/R) within Foundry
- Statistical modeling: regression, classification, validation for readiness/logistics
- Time series forecasting with ARIMA/SARIMA patterns
- Monte Carlo simulation for COA comparison and risk quantification
- Linear programming for resource allocation and scheduling optimization
- Wargame/exercise data collection architecture and aggregation pipelines
- Analytical decision support products (Quiver/Contour) to commander standard
- Communicate uncertainty: confidence intervals, sensitivity analysis, briefing standards

PREREQ: SL 3 • ADVANCED: SL 5G • OPEN PDF →

TM-40H (SL 4H) — AI Engineer Track

SL 3 REQUIRED

AIP Logic, Agent Studio & LLM Integration

AI/ML specialists • 17A/17C

- Author AIP Logic workflows: prompt engineering, chain design, output handling
- Build and configure AIP Agent Studio agents with tools, memory, and orchestration
- Implement LLM integration patterns: ontology data grounding, RAG, context construction
- Apply AI safety requirements: human-in-the-loop gates, output validation, OPSEC
- Write Python transforms that prepare data for AI consumption
- Connect AIP Logic workflows to Object Types and Actions

- Test and red-team AI outputs; evaluate quality against defined standards
- Deploy and monitor AIP Logic workflows in production

PREREQ: SL 3 • ADVANCED: SL 5H • OPEN PDF →

TM-40M (SL 4M) — ML Engineer Track

SL 3 REQUIRED

Code Workspaces, Model Training & Deployment

ML engineers • Data scientists building/deploying models on MSS

- Configure Code Workspaces for model development (GPU, packages, environment management)
- Build and evaluate ML models within the Foundry environment
- Manage model versioning, experiment tracking, and reproducibility
- Deploy models to production and integrate with Ontology Objects and Actions
- Implement MLOps patterns: monitoring, drift detection, retraining triggers
- Apply responsible AI practices and model documentation standards for operational use

PREREQ: SL 3 • ADVANCED: SL 5M • OPEN PDF →

TM-40J (SL 4J) — Program Manager Track

SL 3 REQUIRED

Agile Project Management for Data & AI Capabilities

PMs • Product owners • G8/S8 • Technical team leads

- Stand up Agile project structures (backlog, sprint cadence, ceremonies) for data and AI builds
- Write user stories and acceptance criteria that SL 4G–O developers can execute without ambiguity
- Manage the ML/AI project lifecycle: six phases from Problem Definition through Sustainment
- Translate commander requirements into prioritized, sprint-ready backlogs
- Specify project tracking systems (sprint boards, status dashboards) for SL 4L implementation
- Build and maintain risk registers; manage dependency blockers across specialist tracks
- Conduct production readiness reviews against the Definition of Done before release
- Execute change management plans for new MSS capability rollout to operational units

PREREQ: SL 3 • ADVANCED: SL 5J • OPEN PDF →

TM-40K (SL 4K) — Knowledge Manager Track

SL 3 REQUIRED

Knowledge Repositories, AIP Summarization & Lessons Learned

KMOs • 37F • S2/S3 KM roles • AAR facilitators

- Design knowledge architecture for AAR, lessons learned, doctrine, and SOP repositories
- Build AAR capture systems using Workshop forms and Object Type pipelines
- Design and operate lessons-learned ingestion and tagging pipelines
- Use AIP Logic for knowledge summarization, search augmentation, and theme extraction
- Build full-text and semantic search systems over knowledge repositories
- Manage doctrine and SOP version control within Foundry
- Build personnel expertise mapping (skills/experience registries)
- Design knowledge transfer and unit continuity processes using MSS
- **NEW:** Leverage Document Intelligence (GA) for automated document parsing and extraction
- **NEW:** Use Object Views (GA) for curated knowledge browsing interfaces

PREREQ: SL 3 • ADVANCED: SL 5K • OPEN PDF →

TM-40L (SL 4L) — Software Engineer Track

SL 3 REQUIRED

OSDK, Full-Stack Foundry Applications & Platform SDK

SWEs • 17A/17C • 25D

- Authenticate and query the Foundry Ontology via OSDK (TypeScript/Python)
- Execute Actions, subscribe to Object changes, handle pagination and filtering via OSDK
- Use Foundry Platform SDK for dataset operations, file management, and branch management
- Build TypeScript Functions on Objects (computed properties, bulk query patterns)
- Write and test complex Action validators with TypeScript
- Build Slate applications integrated with the Foundry API
- Apply USAREUR-AF code review and deployment standards for MSS applications
- **NEW:** Use Pilot for AI-assisted code generation within Code Repositories
- **NEW:** Monitor OSDK client health via the Health Dialog dashboard
- **NEW:** Integrate external tools via Model Context Protocol (MCP) connectors

PREREQ: SL 3 • ADVANCED: SL 5L • OPEN PDF →

TM-40N (SL 4N) — UI/UX Designer Track

SL 3 REQUIRED

Soldier Centered Design, Workshop & Slate UI

UI/UX designers • Human factors • GS/contractor designers

- Conduct user research in operational and classified environments (interview, contextual inquiry, usability testing)
- Design information architectures for data-dense operational displays
- Build interactive prototypes from low-fidelity sketches through high-fidelity mockups
- Design Workshop layouts: widget selection, dashboard hierarchy, responsive patterns
- Apply visual design standards for tactical displays: classification marking, contrast, field conditions
- Ensure Section 508 / WCAG 2.1 AA accessibility compliance

PREREQ: SL 3 • ADVANCED: SL 5N • OPEN PDF →

TM-40O (SL 4O) — Platform Engineer Track

SL 3 REQUIRED

Kubernetes, CI/CD, DevSecOps & Infrastructure as Code

Platform engineers • DevOps • SysAdmins • 25D

- Architect and operate Kubernetes clusters for MSS workloads
- Implement Infrastructure as Code with GitOps workflows and continuous reconciliation
- Design CI/CD pipelines: automated build, test, scan, and deploy for MSS applications
- Harden containers using DoD Iron Bank images, vulnerability scanning, and SHA256 digest pinning
- Deploy across classification boundaries and DDIL environments (air-gapped, edge clusters)
- Manage RMF/ATO lifecycle from the infrastructure perspective, STIG compliance
- **NEW:** Configure and optimize Compute Modules (GA) for scalable pipeline execution
- **NEW:** Manage Data Connection source types and third-party integration patterns

PREREQ: SL 3 • ADVANCED: SL 5O • OPEN PDF →

▼ WARFIGHTING FUNCTION TRACKS (SL 4A–F)

DESIGNATION	TRACK	PUBLICATION
SL 4A	Intelligence	TM_40A_INTELLIGENCE.pdf
SL 4B	Fires	TM_40B_FIRES.pdf
SL 4C	Movement & Maneuver	TM_40C_MOVEMENT_MANEUVER.pdf
SL 4D	Sustainment	TM_40D_SUSTAINMENT.pdf

DESIGNATION	TRACK	PUBLICATION
SL 4E	Protection	TM_40E_PROTECTION.pdf
SL 4F	Mission Command	TM_40F_MISSION_COMMAND.pdf

NEXT LEVEL — AFTER SL 4

SL 5 — Advanced Developer Tracks

Expert-level continuation of each SL 4 specialist track. For senior technical leads, platform architects, and developers building enterprise-scale MSS capabilities.

Continue to SL 5 →

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →** For task-level procedures, use the **Task Index →**

SL 5

Advanced Developer Tracks

Prerequisite: SL 4 (by track) • Tracks SL 5G–O • All 8 tracks available

▶ BLUF — ADVANCED TRACKS

The SL 5 series provides advanced-level instruction for each developer track, building directly on the corresponding SL 4 specialist manual (SL 4G–O). Intended for senior technical leads, platform architects, and senior developers designing enterprise-scale MSS capabilities. All 8 tracks are complete and available.

SL 5 SERIES — PUBLICATIONS

TM-50G (SL 5G) — ORSA Advanced

ADVANCED

Advanced ORSA

Prerequisite: SL 4G

- Nonlinear programming, stochastic models
- Agent-based modeling (ABMS)
- Campaign wargame data architecture

TM-50H (SL 5H) — AI Engineer Advanced

ADVANCED

Advanced AI Engineering

Prerequisite: SL 4H

- Multi-agent orchestration & shared state
- Advanced RAG, domain-adapted LLMs
- AI red-team assessment & observability

TM-50M (SL 5M) — ML Engineer Advanced

ADVANCED

Advanced ML Engineering

Prerequisite: SL 4M

- Automated retraining pipelines

- Transformer fine-tuning, GNNs
- Federated retraining, adversarial robustness

TM-50J (SL 5J) — PM Advanced

ADVANCED

Advanced Program Management

Prerequisite: SL 4J

- PI planning, cross-team governance
- GO/SES briefing, Palantir partnership
- Technical debt at program scale

TM-50K (SL 5K) — KM Advanced

ADVANCED

Advanced Knowledge Management

Prerequisite: SL 4K

- Federated KM architecture, NATO integration
- STANAG 4778 conformance
- Knowledge graphs at scale

TM-50L (SL 5L) — SWE Advanced

ADVANCED

Advanced Software Engineering

Prerequisite: SL 4L

- Scale, multi-tenancy, event streaming
- OWASP, SAST, authorized pen testing
- Architecture review, platform governance

TM-50N (SL 5N) — UI/UX Advanced

ADVANCED

Advanced UI/UX Design

Prerequisite: SL 4N

- Design systems at scale, component libraries

- DDIL-aware and cross-domain UI design
- DesignOps, ResearchOps, accessibility at enterprise scale

TM-50O (SL 5O) — Platform Eng Advanced

ADVANCED

Advanced Platform Engineering

Prerequisite: SL 4O

- Multi-cluster fleet management, SRE practices
- RMF/ATO automation, continuous compliance
- Cross-domain infrastructure, developer experience engineering

• NOTE — PREREQUISITES

Each SL 5 track requires completion of the corresponding SL 4 track. Personnel should confirm SL 4 proficiency with their data steward before beginning SL 5 content. Contact the USAREUR-AF Operational Data Team for access questions.

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →** For task-level procedures, use the **Task Index →**

DRAFT PUBS

Draft Data Literacy Publications

DRAFT — Not yet approved for distribution • Foundational data literacy content • Platform-agnostic

⚑ CAUTION — DRAFT PUBLICATIONS

These publications are in draft status and have not been approved for official distribution. Do not distribute outside the USAREUR-AF training program without authorization from the USAREUR-AF Operational Data Team.

▸ BLUF

Before touching MSS, understand *why* data matters. Data Literacy for Senior Leaders is written for commanders and senior leaders. Data Literacy Technical Reference is the comprehensive reference for all personnel. Neither publication is platform-specific.

SENIOR LEADERS (O-5+ / SGM+)

Data Literacy for Senior Leaders

Written for commanders, senior NCOs, and senior Civilians. Covers command responsibilities, evaluating data products, directing a data-capable formation, and decision frameworks.

Format: Short (~20–30pp). Principles, not procedures. Chapter/paragraph numbered.

Key topics: Commander's data responsibilities • Evaluating analytical products • Standing up MSS capability • Data governance and stewardship

ALL PERSONNEL

Data Literacy Reference

Comprehensive platform-agnostic data literacy reference. Recommended prior reading before SL 1.

Format: Long (~50–100pp). Examples, vignettes, detailed explanations, annexes.

Key topics: Data types and structures • Pipeline concepts • Data quality • Analysis fundamentals • Security and classification • Operational data integration • Governance

FULL PUBLICATIONS INDEX

PUBLICATION	AUDIENCE	PURPOSE	WHEN TO READ
FOUNDATION — ALL PERSONNEL			
Data Literacy (SL)	O-5+ / SGM+, Sr Civilians	Principles, command responsibilities	Before directing MSS use
Data Literacy	All personnel	Comprehensive data literacy reference	Before SL 1 (recommended)
SL 1	All personnel	Operate MSS as end user	Before first MSS access
SL 2	All staff	Build pipelines, Ontology, Workshop via UI — no code	After SL 1
SL 3	Data-adjacent specialists	Design complex apps; governance; C2DAO standards	After SL 1 + SL 2
SL 4 — WARFIGHTING FUNCTION TRACKS (BY WFF ASSIGNMENT)			
SL 4A	G2/S2, MI, ISR	Intelligence WFF MSS integration	After SL 3
SL 4B	FA, fire support	Fires WFF MSS integration	After SL 3
SL 4C	Maneuver, G3/S3	Movement & Maneuver WFF MSS integration	After SL 3
SL 4D	G4/S4, logistics	Sustainment WFF MSS integration	After SL 3
SL 4E	Air defense, CBRN, force protection	Protection WFF MSS integration	After SL 3
SL 4F	G6/S6, C2, networks	Mission Command WFF MSS integration	After SL 3
SL 4 — TECHNICAL SPECIALIST TRACKS (BY ROLE/MOS)			
SL 4G	ORSA / FA49	Statistical modeling, simulation, wargame analytics	After SL 3

PUBLICATION	AUDIENCE	PURPOSE	WHEN TO READ
SL 4H	AI Engineers	AIP Logic authoring, Agent Studio, LLM integration	After SL 3
SL 4M	ML Engineers	Code Workspaces, model training, MLOps	After SL 3
SL 4J	PMs / G8	PM dashboards, milestone tracking, portfolio analysis	After SL 3
SL 4K	KMs / KMOs	Knowledge repositories, AIP summarization, lessons learned	After SL 3
SL 4L	SWEs	OSDK, full-stack Foundry apps, TypeScript Functions	After SL 3
SL 4N	UI/UX Designers	Soldier Centered Design, Workshop & Slate UI, accessibility	After SL 3
SL 4O	Platform Engineers	Kubernetes, CI/CD, DevSecOps, Infrastructure as Code	After SL 3
SL 5 — ADVANCED TECHNICAL TRACKS (BY ROLE/MOS)			
SL 5G–O	Senior developers (all tracks)	Advanced versions of each SL 4 specialist track	After SL 4 (by track)
CDA REFERENCE — SL 3 AND SPECIALIST TRACKS (SL 4G–O)			
EA Series (00–05)	SL 3+, SL 4K, SL 4L	Enterprise Architecture — foundation, schools of thought, artifacts, governance, military application	With or after SL 3
CDA Doctrine Overview	SL 4G–O	Doctrine-driven development; JRTC lessons; Foundry Ontology blueprint	At start of SL 4G–O
Identity vs. Classification	SL 3, SL 4K, SL 4L	Identity resolution and classification governance at scale	With SL 3
Enterprise Data	SL 4J, SL 4K	Authoritative data architecture, ontology, and semantic governance	With SL 4J/K

PUBLICATION	AUDIENCE	PURPOSE	WHEN TO READ
Compass		standard	
CDA Slide Library	All tracks (prereq reading)	29 decks — Intro To Data (SL 1 prereq), Data 101 (SL 2 prereq), Data 201 (SL 3/40G–O prereq)	Before each TM level

CDA REFERENCE MATERIAL

• SPECIALIST TRACK PREREQ READING

The CDA Reference material below is required reading for SL 3 and all SL 4G–O specialist tracks. It is **not** required for SL 1/20 or WFF tracks (SL 4A–F). Consult the DEPENDENCY_MAP for per-track required reading.

ENTERPRISE ARCHITECTURE SERIES

EA-00 through EA-05

Six-module reference series covering EA foundations, schools of thought, artifacts and views, governance, and military application. Supports SL 3, SL 4K, and SL 4L.

Key topics: EA vs DA • TOGAF, Zachman, DODAF frameworks • Capability mapping • NAF/ArchiMate • Army EA governance

CDA DOCTRINE SERIES

Doctrine-Driven Development

Doctrine-aligned data product design using JRTC lessons learned. Covers doctrine-first Ontology design, the Three-Generation Dilemma, and AVT25 assessment case study.

Key topics: MDMP data support • COA analysis modeling • Doctrinal object types • AVT25 tools case study

CASE STUDY

XVIII Airborne Corps — Fighting with Live Data

XVIII ABC's corps-level ODT pilot. Their organizational journey, manning structure (PM + UX + SWE + DE + DS), problem-solution development process, Program Increment cycles, and BDA

visualization case study (prototype → MVP → POR in 9 months). *Military Review*, Feb 2026.

Key topics: ODT organization • TIO governance • ASWF methodology • Exercise integration • Echeloned ODT employment • Decision optimization

Required reading: SL 4J, SL 4F • **Recommended:** All SL 4 tracks

CASE STUDY

Lessons Learned — AVT25 Assessment Tools

How five tools in the same enclave multiplied work exponentially by failing to share doctrinal primitives. The case for doctrine-first shared data architecture.

Key topics: Shared primitives • DBO architecture • Exponential work multiplication • Time cost analysis

REFERENCE

Achieving Decision Dominance (Adkins)

One officer's thought piece proposing terminology for decision optimization at echelon. Introduces useful shorthand: operationalized data, Automated Fighting Products (AFP), and Decision Optimization Teams. Names the Maven Smart System as an ASCC-level COP platform. *Military Review*, Jan–Feb 2025.

Key topics: Operationalized data • AFP evolution • DOT at echelon • FA 26B/49/57 workforce • Training pipeline

Supplementary reading: SL 4F, SL 4G, SL 4J • **Recommended:** All SL 4 tracks, Senior Leaders

EXTERNAL DOCTRINE & INSTITUTIONAL SOURCES

• INSTITUTIONAL REFERENCES

The following sources from MCCoE, CALL, and TRADOC inform MSS curriculum design and are recommended supplementary reading for instructors and course developers.

MCCOE

Mission Command Center of Excellence — Decision Optimization CONOPS

MCCoE's conceptual framework for integrating data-centric capabilities into mission command. Defines decision optimization at echelon and the role of Operational Data Teams in the command post.

Relevance: SL 4F (Mission Command), SL 4J (Program Manager), EXEC (Senior Leader)

CALL

Center for Army Lessons Learned — FY24 MCTP Key Observations

CTC observer trends on data-centric operations, common gaps in unit-level data readiness, and recommendations for training programs. Published Feb 2025.

Relevance: SL 1 (all personnel), SL 4A–F (WFF tracks), instructor development

TRADOC

TRADOC Data Literacy Framework

TRADOC's institutional approach to data literacy across the force. Defines competency levels, assessment criteria, and integration with PME. Informs the MSS SL 1/20 foundation sequence.

Relevance: Data Literacy publications, SL 1, SL 2, instructor certification (T3-I)

MCCOE / ARMY.MIL

Army's Combined Arms Command to Integrate Maven C2 Smart System

Public announcement of Maven C2 integration into training and education at the Combined Arms Center. Establishes institutional backing for MSS-based training programs. Published Feb 2026.

Relevance: All tracks — institutional context for the MSS training program

CORE DATA LITERACY CONCEPTS

DATA TYPES

Structured (tables, rows, columns) • Semi-structured (JSON, XML) • Unstructured

DATA PIPELINE

Raw → Staging → Curated → Ontology → Application. Each layer adds quality,

(documents, images). MSS ingests all three types from Army source systems.

structure, and meaning. Never modify raw data — it is the source of truth.

DATA QUALITY

Accuracy, completeness, consistency, timeliness, uniqueness. Bad decisions follow bad data. Report quality issues to your data steward rather than working around them.

DATA GOVERNANCE

Ownership, stewardship, access control, classification markings. Every dataset has an owner. Every user has a role. Access is audited. Misuse is tracked and prosecuted.

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →** For task-level procedures, use the **Task Index →**

MSS TRAINING HUB

MAVEN SMART SYSTEM — USAREUR-AF

Version 3.1 • March 2026

◆ NEW TO MSS?

Start with the Quick Start guide before reading Skill Level (SL) 1. Operational in 30 minutes: log in via CAC, navigate to your unit's app, filter data, export a view. → **QUICK_START.pdf** • No account yet? Contact your unit data steward.

▶ BLUF

The MSS curriculum is organized by Skill Level (SL) per DA PAM 611-21. All personnel start at SL 1. Builders add SL 2; data-adjacent specialists continue to SL 3; technical roles select their SL 4 track. Use "Find My Track" below or browse the reference table.

⚡ Find My Track

— answer two questions to get your personalized training path ▲

1 WHAT BEST DESCRIBES YOUR ROLE?

📊
Data Consumer
I view dashboards, reports, and data in MSS.

⭐
Senior Leader
O-5+ / SGM+ directing a data-capable formation.

🛠
No-Code Builder
I build dashboards, forms, or pipelines — no code.

🧩
Advanced Builder
Complex apps, ontology architecture, governance. Fork point for SL 4.

🎖
Warfighting Function
Go deeper in your warfighting function (Intel, Fires, Maneuver, etc.) after SL 3.

💻
Technical Specialist
ORSA, AI/ML, Knowledge Mgmt, Software, UX, or Platform Engineering — coding and engineering roles after SL 3.

📖 Find Your Manual

— quick-reference table by role ▼

TRAINING PATH

CORE PROGRESSION (SKILL LEVELS)

FOUNDATION

FOUNDATION

SL 1
Maven User
8 hrs • Unit-level

SL 2
Builder
40 hrs • 5 days

SL 3
Advanced Builder
40 hrs • 5 days

SL 4/5
Specialist (WFF/Tech)
14 tracks • 40 hrs ea.

PARALLEL TRACKS

Bootcamp
Foundry Bootcamp
5 days • 2x Quarterly
Prereq: SL 2

EXEC
Senior Leader
1 day • O-5 / E-9+
No prereq

T3
Instructor Pipeline
T3-F (UDT) • T3-I (Cert)
Prereq: SL 2 / SL 3

START YOUR TRAINING

Foundation — SL 1 → SL 2 → SL 3

All personnel begin here. SL 1 gets you operational; SL 2 builds no-code skills; SL 3 unlocks specialist tracks.

Start with SL 1 →

• MSS ACCOUNT ACCESS

MSS access requires a provisioned account. Submit your request through your unit data steward or at **mss.data.mil**. Provisioning generally completes within 24 hours; if access is not active after 24 hours, contact your data steward directly.

• NOT FINDING WHAT YOU NEED?

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SUPPORT

Getting Help — USAREUR-AF

▶ BLUF

Know who to call before you have a problem. Route issues correctly from the start. Collect information before calling for help — it speeds resolution significantly.

⚠ WARNING — SECURITY INCIDENTS

If you suspect a security violation, report immediately to your supervisor and unit security officer. Do not investigate or resolve it yourself. Preserve the screen state; do not close the window or clear the browser.

CONTACT ROUTING

ISSUE	ROUTE TO	PRIORITY
Cannot log in / CAC issues	MSS Help Desk	Normal
No access to a project or dataset	Unit data steward	Normal
Data appears incorrect	Unit data steward (do not self-correct)	Normal
System error, crash, or outage	MSS Help Desk + screenshot + error code	Normal
Application broken or not loading	MSS Help Desk	Normal
Building question / how-to	Unit data lead or USAREUR-AF data team	Normal
Governance / C2DAO question	USAREUR-AF C2DAO	Normal
Security incident	**Supervisor + unit security officer**	**IMMEDIATE**

COMMON DAILY TASKS

TASK	HOW
Find a record	Search bar or Filter Panel
Filter the view	Select values in the Filter Panel

TASK	HOW
Export data	**Export** / **Download** button → CSV or Excel
Submit or update a record	Click record → **Action button** → fill in → Submit

WHEN IT BREAKS

CAN'T LOG IN

Check CAC is fully inserted; try a different port. No account? Request at **mss.data.mil** or through your data steward. Provisioning generally within 24 hrs; if not active after 24 hrs, contact your data steward.

APP WON'T LOAD

Hard-refresh (Ctrl+Shift+R). Clear cache. Try a different browser.

BUTTON GREYED OUT / NO ACCESS

You're missing a role or write permission. Contact your data steward to request the correct access level.

SECURITY — DO NOT

- Do **not** export data to a personal device or unapproved storage
- Do **not** share your MSS **credentials** with anyone — URLs and screenshots are fine unless data is sensitive
- Do **not** enter classified information into MSS unless your instance is approved for that classification level
- Do **not** screenshot or share MSS screens containing data above your network's approved classification
- Do **not** use MSS on public or unsecured Wi-Fi
- If you see data you should not have access to — **stop and report to your data steward immediately**

BEFORE CALLING FOR HELP — COLLECT THIS INFORMATION

- Your username and unit

- Name of the application, dataset, or pipeline you were using
- Exact error message (screenshot preferred)
- Time the error occurred (local or Zulu — state which)
- Steps that led to the error in order
- Browser and workstation you are using

PREREQUISITES BEFORE FIRST LOGIN

1. **Annual Cyber Awareness Training** — required for all DoD personnel; must be current
2. **MSS User Onboarding Brief** — provided by unit data steward
3. **Account request approved** — submit at **mss.data.mil** or through your unit data steward; provisioning generally completes within 24 hours. If access is not active after 24 hours, contact your data steward directly.

IMPORTANT: If you work on multiple enclaves (NIPR, SIPR, MPE, etc.), you must complete account setup and first login on **each enclave separately**. Your account on one enclave does not carry over to another.

USAREUR-AF DATA TEAM

LOCATION & HIGHER HQ

EUCOM Theater — Europe & Africa

Headquarters, United States Army Europe and Africa
USAREUR-AF Operational Data Team
Army AI/Data Accelerator (C2DAO)

PUBLICATIONS

All training publications are maintained by the USAREUR-AF Operational Data Team. Contact your unit data steward for access to source files and this application. Version 3.1 — March 2026.

FEEDBACK & CORRECTIONS

Route corrections through your unit data steward to the USAREUR-AF Operational Data Team. Include: publication name, section, and description of the issue.

• NOTE — DISTRIBUTION

These are DRAFT documents — not yet approved for distribution. Do not distribute outside your organization without consulting your data steward.

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →** For task-level procedures, use the **Task Index →**

DOCUMENTS

All Training Publications

Click any publication to open the PDF

SEARCH: Filter publications by title, track, or keyword...

Reference Documents

DATA LIT — SENIOR LEADERS

Data Literacy for Leaders

Senior leader version — strategic principles, decision frameworks. Commanders and O6+.

DATA LITERACY

Data Literacy — Comprehensive Reference

Full data literacy reference, examples, vignettes, and annexes. Platform-agnostic.

GLOSSARY

Data & Foundry Term Glossary

Combined data concepts and Foundry/MSS term equivalency. v1.3.

STANDARDS

Naming & Governance Standards

Codified naming conventions, governance checklists — 10 sections, 2 appendices.

QUICK REF

MSS Quick Reference Cheatsheet

At-a-glance reference card for common MSS tasks and patterns.

MTP

Mission Training Plan — MSS

TLOs, ELOs, Go/No-Go criteria, schedule templates. 961 lines.

Technical Manuals — Foundation (All Staff)

TM-10 (SL 1)

Maven User

Operate MSS, consume data, use dashboards, Gaia, command-level applications, and AIP tools embedded in Workshop. Required for all staff.

TM-20 (SL 2)

No-Code Builder

Solution Designer, Pipeline Builder, Workshop, Ontology (Interfaces, Object Views, Actions). All staff with build responsibilities.

TM-30 (SL 3)

Advanced No-Code Builder

Advanced Workshop (Kairos, Target Workbench), complex pipelines, AIP Logic, basic Agent Studio, Automations, Machinery. Data-adjacent MOS (17/25-series, G2).

▶ TM-40 WARFIGHTING FUNCTION TRACKS (6 — CLICK TO EXPAND)

▶ TM-40 TECHNICAL SPECIALIST TRACKS (8 — CLICK TO EXPAND)

▶ TM-50 ADVANCED TECHNICAL TRACKS (8 — CLICK TO EXPAND)

▶ TRAIN THE TRAINER — T3 (2 COURSES + SOPS — CLICK TO EXPAND)

▶ CONCEPTS GUIDES (23 — CLICK TO EXPAND)

▶ PRACTICAL EXERCISES (13 — CLICK TO EXPAND)

▶ PRE-ASSESSMENT TESTS (27 — CLICK TO EXPAND)

▶ POST-ASSESSMENT TESTS (26 — CLICK TO EXPAND)

▶ COURSE SYLLABI (26 — CLICK TO EXPAND)

▶ SELF-STUDY GUIDES (17 — CLICK TO EXPAND)

▶ LESSON PLANS (5 — CLICK TO EXPAND)

▶ ADMINISTRATIVE & INSTITUTIONAL (12 — CLICK TO EXPAND)

▶ ENTERPRISE ARCHITECTURE SERIES (6 — CLICK TO EXPAND)

▶ ARCHITECTURE & DESIGN REFERENCES (28 — CLICK TO EXPAND)

▼ EXTERNAL DOCTRINAL & STRATEGIC REFERENCES

◆ REFERENCE ONLY

Items below are **not clickable links**. They are external publications listed here for reference and citation. Obtain them through official Army, DoD, or NATO distribution channels.

• CLASSIFICATION

References are categorized as **Doctrine** (regulatory authority — ARs, FMs, DoD Directives, NATO STANAGs) or **Strategic Guidance** (authoritative but not regulatory — strategies, plans, frameworks). This distinction matters for compliance and citation.

ARMY DOCTRINE & REGULATION

PUBLICATION	TITLE	TYPE	TRACKS
ADP 3-0	Operations	Doctrine	WFF (A–F)
ADP 3-19	Fires	Doctrine	SL 4B
ADP 3-37	Protection of the Force (Jul 2019)	Doctrine	SL 4E
ADP 3-90	Offense and Defense	Doctrine	SL 4C
ADP 5-0	The Operations Process	Doctrine	SL 4F, SL 4G

PUBLICATION	TITLE	TYPE	TRACKS
ADP 6-0	Mission Command (Jul 2019)	Doctrine	SL 4F
ADP 7-0	Training	Doctrine	Training Mgmt
AR 25-1	Army Information Technology (Jul 2019)	Regulation	All
AR 25-2	Army Cybersecurity (Apr 2019)	Regulation	SL 4H, SL 4M, SL 4L
AR 25-30	Army Publishing Program	Regulation	SL 5H
AR 25-400-2	Army Records Management	Regulation	SL 4K
AR 5-11	Management of Army Models and Simulations	Regulation	SL 4G
AR 71-9	Warfighting Analysis	Regulation	SL 4G
AR 350-1	Army Training and Leader Development	Regulation	Training Mgmt
AR 525-2	Force Protection	Regulation	SL 4E
AR 530-1	Operations Security	Regulation	SL 4E
FM 2-0	Intelligence (Oct 2023)	Doctrine	SL 4A
FM 3-0	Operations (Mar 2025)	Doctrine	WFF (A–F)
FM 3-01	U.S. Army Air and Missile Defense	Doctrine	SL 4B
FM 3-09	Fire Support and Field Artillery Operations (Aug 2024)	Doctrine	SL 4B, SL 4C
FM 3-12	Cyberspace and EW Operations	Doctrine	SL 4E, SL 4H, SL 4M, SL 4L
FM 3-27	Army Global Ballistic Missile Defense	Doctrine	SL 4B

PUBLICATION	TITLE	TYPE	TRACKS
FM 3-55	Information Collection	Doctrine	SL 4A
FM 3-60	Targeting (Aug 2023)	Doctrine	SL 4B
FM 3-81	Maneuver Enhancement Brigade	Doctrine	SL 4C
FM 3-90	Offense and Defense (May 2023)	Doctrine	SL 4C
FM 4-0	Sustainment (Aug 2024)	Doctrine	SL 4D
FM 1-0	Human Resources Support	Doctrine	SL 4D
FM 5-0	Planning and Orders Production (Nov 2024)	Doctrine	SL 4F, SL 4C
FM 6-0	Commander's Activities (May 2022)	Doctrine	SL 4F, SL 4C
FM 7-0	Training (Jun 2021)	Doctrine	Training Mgmt
ATP 2-01	Collection Management (May 2023)	Doctrine	SL 4A
ATP 2-33.4	Intelligence Analysis	Doctrine	SL 4A
ATP 2-22.9-1	PAI/OSINT (Oct 2023)	Doctrine	SL 4A
ATP 3-01.81	Counter-UAS	Doctrine	SL 4B
ATP 3-09.42	Fire Support for M&M	Doctrine	SL 4B
ATP 3-13.3	Army Operations Security	Doctrine	SL 4E
ATP 3-90.4	Combined Arms Mobility	Doctrine	SL 4C
ATP 5-0.1	Army Design Methodology	Doctrine	SL 4F

PUBLICATION	TITLE	TYPE	TRACKS
ATP 5-0.3	Multi-Service Tactics for Ops Assessment	Doctrine	SL 4G
ATP 6-01.1	Techniques for Effective Knowledge Management	Doctrine	SL 4K, SL 5K
TC 6-0.2	Battle Staff Operations	Doctrine	SL 4F
DA PAM 5-11	Verification, Validation & Accreditation	Doctrine	SL 4G
DA PAM 25-1-1	IT Implementation Instructions	Doctrine	SL 4K, SL 4L
DA PAM 25-2-5	Cybersecurity Technical Reference	Doctrine	SL 4H, SL 4M, SL 4L
DA PAM 25-40	Army Publishing Program Procedures	Doctrine	Standards
DA PAM 25-403	Army Records Information Management	Doctrine	SL 4K
DA PAM 600-3	Officer Professional Development	Doctrine	SL 4G

DoD DIRECTIVES & INSTRUCTIONS

PUBLICATION	TITLE	TYPE	TRACKS
DoDD 3000.09	Autonomy in Weapon Systems (Jan 2023)	Directive	WFF (A–F)
DoDI 5000.87	Software Acquisition Pathway (Oct 2020)	Instruction	SL 4L, SL 5L, SL 4J, SL 5J
Army Directive 2024-02	Agile Software Dev & Acquisition (Dec 2024)	Directive	SL 4L, SL 5L, SL 4J, SL 5J
Army Directive 2024-03	Army Digital Engineering	Directive	SL 4H, SL 4M, SL 4L

TRADOC PUBLICATIONS

Published at adminpubs.tradoc.army.mil, not armypubs.army.mil

PUBLICATION	TITLE	TYPE	TRACKS
TR 350-70	Army Learning Policy and Systems	Regulation	Training Mgmt
TP 350-70-3	Faculty and Staff Development Program	Pamphlet	Training Mgmt
TP 350-70-7	Army Educational Processes	Pamphlet	Training Mgmt
TP 350-70-14	Training Development in Institutional Domain	Pamphlet	Training Mgmt

NATO STANDARDS & AGREEMENTS

PUBLICATION	TITLE	TYPE	TRACKS
ADatP-34 / NISP	C3 Interoperability Standards and Profiles	Standard	SL 4K, SL 4L
STANAG 5636 / NCMS	Core Metadata Specification	STANAG	SL 4K, SL 5K
STANAG 5643 (proposed)	MIM Governance Standard	STANAG	SL 4K, SL 4L, SL 5K, SL 5L
ADatP-5644	Web Service Messaging Profile (WSMP)	Standard	SL 4L, SL 5L
ADatP-36	Friendly Force Information (FFI)	Standard	SL 4A, SL 4C
STANAG 5527	FFT Systems Interoperability	STANAG	SL 4A

DoD & ARMY STRATEGIC GUIDANCE (not doctrine)

DOCUMENT	AUTHORITY	DATE	TRACKS
DoD Data Strategy	OSD	Oct 2020	All

DOCUMENT	AUTHORITY	DATE	TRACKS
DoD Data, Analytics & AI Adoption Strategy	CDAO	Nov 2023	All
DoD Responsible AI Strategy	CDAO	Jun 2024	SL 4H/M, SL 5H/M
DoD Zero Trust Reference Architecture v2.0	DISA/NSA	Jul 2022	SL 3
DoD AI Cybersecurity Risk Mgmt Guide	DoD CIO	2024	SL 4H/M, SL 5H/M
DoD Software Modernization Strategy	OSD CIO	Feb 2022	SL 4L, SL 5L
JADC2 Strategy Summary	Joint Staff	Mar 2022	WFF (A–F), SL 4G
JCOIE	Joint Staff J-7	Current	SL 4F
Army Data Plan	Army CIO	Oct 2022	All
Army Cloud Plan	Army CIO	Oct 2022	SL 1, SL 2, SL 3
UDRA v1.1	DASA(DES)	Feb 2025	SL 3, Specialist (G–O)
Army CIO Data Stewardship Memo	Army CIO	Apr 2024	SL 1, SL 2, SL 3, SL 4K

NATO STRATEGIC GUIDANCE (not doctrine)

DOCUMENT	DATE	TRACKS
NATO Data Strategy for the Alliance	Feb 2025	SL 3, SL 4K, SL 5K
NATO Data Centric Reference Architecture v2	2025	SL 3
NATO Data Quality Framework for the Alliance	Aug 2025	SL 3
NATO Digital Transformation Implementation Strategy	Oct 2024	WFF (A–F)

DOCUMENT	DATE	TRACKS
NATO Warfighting Capstone Concept	2021	SL 4F

▼ PROFESSIONAL READING & LESSONS LEARNED (65+ ARTICLES)

◆ REFERENCE ONLY

Items below are **not clickable links**. They are curated articles from Army professional journals, military publications, and think tanks. Obtain them through the publishing organization or your unit library. Full reading lists are included as appendices in each TM publication.

MILITARY REVIEW — Army University Press (14)

TITLE	DATE	TRACKS
Data-Centric at the Division: 3ID's One-Year Journey to Transform and Modernize	Jan 2025	All
Modernizing Military Decision-Making: Integrating AI into Army Planning	Aug 2025	SL 4F, SL 4H, SL 4G
The Military Needs Frontier Models	Aug 2025	SL 4H, SL 4M, SL 4L
Exploring AI-Enhanced Cyber and Information Operations Integration	Mar-Apr 2025	SL 4E, SL 4A, SL 4H
Authorities and the Multidomain Task Force	Mar-Apr 2025	SL 4A, SL 4B, SL 4F
Taking a Data-Centric Approach to Unit Readiness	2024	All, esp. SL 4G
Attaining Readiness by Developing a Data-Centric Culture	2024	All, esp. SL 4J

TITLE	DATE	TRACKS
Sustaining Our People Advantage in Data-Centric Warfare	2024	All
AI as a Combat Multiplier: Using AI to Unburden Army Staffs	Sep 2024	SL 4H, SL 4F, SL 4G
Transforming the Multidomain Battlefield with AI	2024	SL 4H, SL 4M, SL 4A
The Coming Military AI Revolution	May-Jun 2024	SL 4H, SL 4M
AI in Modern Warfare: Strategic Innovation and Emerging Risks	Sep-Oct 2024	All
Advancing Counter-UAS Mission Command Systems	May-Jun 2024	SL 4E, SL 4F
The True Test of Mission Command	Sep-Oct 2024	SL 4F

PARAMETERS — Army War College Quarterly (3)

TITLE	DATE	TRACKS
Responsibly Pursuing Generative AI for the War Fighter	Winter 2025-26	SL 4H, SL 4M, All
Integrating AI and ML into COP and COA Development	2024-25	SL 4G, SL 4H, SL 4F
Trusting AI: Integrating AI into the Army's Professional Ethic	2024	All

MIPB — Military Intelligence Professional Bulletin (6)

TITLE	DATE	TRACKS
FRIDAY: Unlocking OSINT for a Data-Driven Army	2025	SL 4A, SL 4H, SL 4L

TITLE	DATE	TRACKS
Intelligence Support to Information Advantage	Jan-Jun 2026	SL 4A, SL 4K
Using AI to Create Digital Enemy Commanders	Jul-Dec 2025	SL 4H, SL 4M, SL 4A
The Market Knows Best: Prediction Markets for National Security	Jul-Dec 2025	SL 4A, SL 4G
Army Transitioning to Support Deep Sensing in MDO	Jul-Dec 2025	SL 4A, SL 4B, SL 4C
Open-Source Intelligence Support to Targeting	2024	SL 4A, SL 4B

FIELD ARTILLERY BULLETIN — Line of Departure (6)

TITLE	DATE	TRACKS
The New Digital Kill Chain	2025	SL 4B, SL 4L
AI's New Frontier in War Planning	2025	SL 4B, SL 4H
Project Convergence: Revolutionizing Targeting in LSCO	2025	SL 4B, SL 4A, SL 4G
Enhancing Tactical Level Targeting With AI	2024	SL 4B, SL 4H, SL 4M
The Future of Strategic Fires Target Acquisition	2024	SL 4B, SL 4A
The Combat Aviation Brigade and Digital Call for Fire	2024	SL 4B, SL 4C

NCO JOURNAL — Army University Press (3)

TITLE	DATE	TRACKS
Knowledge Management and The Old Guard	Aug 2025	SL 4K, SL 4F
From Data to Wisdom	Feb 2025	All

TITLE	DATE	TRACKS
Artificial Intelligence and Future Warfare	Sep 2025	SL 4H, SL 4M, All

ARMY SUSTAINMENT — Army Logistics University (4)

TITLE	DATE	TRACKS
Army Sustainment Enterprise's Delayed Approach to Data Modernization	Winter 2025	SL 4D, SL 4K
Predictive Logistics: Reimagining Sustainment on the 2040 Battlefield	Winter 2025	SL 4D, SL 4H, SL 4M, SL 4G
Enabling Logistics in Contested Environments	Spring 2025	SL 4D, SL 4G
Advancing to Data-Driven Logistics Operations	2024	SL 4D, SL 4K

ARMY AL&T MAGAZINE (7)

TITLE	DATE	TRACKS
Accelerating the Army's AI Strategy	2024-25	SL 4H, SL 4J, All
Commoditizing AI/ML Models	2024-25	SL 4H, SL 4M, SL 4L
The Army's Data (Ad)Vantage	2024	All
The Software Advantage	2024-25	SL 4L, SL 4J
Army Intelligence	2025	SL 4A, SL 4H
Emerging Technology and Modernizing the Army	2024-25	All
Reality Check (AI/ML implementation)	2024-25	SL 4H, SL 4M, SL 4J

ARMY COMMUNICATOR — Cyber CoE (3)

TITLE	DATE	TRACKS
Leading in Data Centricity, C2 Fix Best Practices	Spring 2025	SL 4E, SL 4F, SL 4L
Army Communicator Spring 2024	Spring 2024	SL 4E, SL 4L
Army Communicator January 2024 — ITN Suite	Jan 2024	SL 4E, SL 4C

FROM THE GREEN NOTEBOOK (3)

TITLE	DATE	TRACKS
How To Be a Data Literate Leader — And Why It Matters	Mar 2024	All, SL 4K
Harnessing the Power of Knowledge Management	Apr 2024	SL 4K, SL 4F
Understanding Weapons of Math Destruction	Jul 2024	SL 4G, SL 4H, SL 4M

INFANTRY MAGAZINE — Maneuver CoE (1)

TITLE	DATE	TRACKS
Moneyball for Gunnery — 1/4 ID BCT data analytics	2024	SL 4C, SL 4G

SMALL WARS JOURNAL (4)

TITLE	DATE	TRACKS
Data as Firepower: Data Superiority as a Warfighting Concept	Aug 2025	All
Elevating Information: Why the Army Should Establish Information as a Core WfF	Apr 2025	SL 4A, SL 4F, SL 4K
Accelerating Decision-Making: Integrating AI into the Modern Wargame	Feb 2026	SL 4G, SL 4H, SL 4F
AI-Enabled Wargaming at CGSC	Jan 2026	SL 4G, SL 4H, SL 4F

WAR ON THE ROCKS (1)

TITLE	DATE	TRACKS
The U.S. Army, AI, and Mission Command	Mar 2025	SL 4F, SL 4H

MODERN WAR INSTITUTE — West Point (1)

TITLE	DATE	TRACKS
Leadership, Lethality, and Data Literacy	2024	All

CALL — Center for Army Lessons Learned (1)

TITLE	DATE	TRACKS
FY24 MCTP Key Observations	Feb 2025	All

CDA Slide Decks — Conceptual Prereqs

◆ RECOMMENDED READING

The CDA slide decks provide conceptual grounding that the TM series assumes but does not teach. They are recommended reading, not required prerequisites. Organized below by the TM level they support.

▶ SL 1 — ORIENTATION (2 DECKS — CLICK TO EXPAND)

▶ SL 2 — INTRO TO DATA (10 DECKS — CLICK TO EXPAND)

▶ SL 2 / SL 3 — DATA 101 (4 DECKS — CLICK TO EXPAND)

▶ SL 3 — ADVANCED (8 DECKS — CLICK TO EXPAND)

▶ SL 3 / SL 4 — DATA 201 (6 DECKS — CLICK TO EXPAND)

▶ SL 4 — SPECIALIST TRACK DECKS (4 DECKS — CLICK TO EXPAND)

▶ SL 5 — ADVANCED TRACK DECKS (6 DECKS — CLICK TO EXPAND)

▶ PROGRAM & BRIEFING DECKS (3 DECKS — CLICK TO EXPAND)

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →** For task-level procedures, use the **Task Index →**

TASK INDEX

What Do You Want To Do?

Search • Filter by category • Click badge to open PDF

MSS REF

MSS TASK REFERENCE INDEX

USAREUR-AF Operational Data Team | What do you want to do? Search below.

SEARCH: Type anything — e.g. spreadsheet, chart, lo

518 / 315

FILTER: All | Setup & Access | Navigation | Data & Analysis | No-Code Building | WFF Tracks (40A–F) | Ontology & Modeling | AIP & AI | Code & SDK | Machine Learning | Program Mgmt | Knowledge Mgmt | Governance & Security | Advanced / Expert | T3 (Train-the-Trainer)

SETUP & ACCESS

Set up account before first login	SL 1	Ch 2
Log in to MSS with CAC	SL 1	Ch 2
Log out of MSS	SL 1	Ch 2
Locate self-paced training resources	SL 1	Ch 1
Verify your own markings and access level	SL 1	Ch 7
Configure new ORSA Code Workspace	SL 4G	Ch 2
Create and configure Code Workspace	SL 4M	Ch 2
Manage packages in Code Workspace	SL 4G	

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DASHBOARDS

Analytics & Operations Suite

Streamlit applications • Click any card to open

▶ BLUF

These dashboards provide real-time analytics, operational management, and content quality tools for the MSS training program. Click any card below to load the application. On the ODT local network, apps load live. On Cloudflare, apps require VPN to the ODT network.

TRAINING ANALYTICS

📋 Readiness Tracker :8501

Soldier training timelines, overdue flagging, printable records.

📊 Exam Analytics :8502

Score distributions, cohort comparison, item discrimination, question improvement.

📝 AAR Aggregator :8503

AAR trend analysis, priority matrix, keyword extraction, GO/NO-GO tracking.

📈 Training Metrics :8511

Executive dashboard — aggregates all training data for CG/DCG briefings.

TRAINING OPERATIONS

🎯 Progress Tracker :8504

Individual Soldier progress, goal tracking, stalled-Soldier alerts.

📅 MTT Scheduler :8505

Mobile Training Team scheduling, resource allocation, calendar view.

👤 Enrollment Manager :8508

Class enrollment, waitlists, rosters, and seat management.

🧑 Instructor Manager :8512

Instructor assignments, certifications, and availability tracking.

CONTENT & QUALITY

✅ Data Quality :8510

Pipeline health monitoring, metric trending, active alerts.

🔗 XRef Validator :8506

Cross-reference validation — find broken links and stale references.

📚 Curriculum Tracker :8513

🔍 Glossary Search :8507

Document version control, review cycles, freshness tracking.

Full-text search across the MSS data foundry glossary.

💡 Lessons Learned :8514

Lessons learned database — search, tag, and trend analysis.

DISTRIBUTION & SYNC

📦 Offline Packager :8509

Build offline content packages for disconnected environments.

☁ SharePoint Sync :8515

Sync training content to SharePoint for enterprise distribution.

• ACCESS

Dashboards run on the ODT local network. To access remotely, connect via VPN first. Each application is available at `http://<host>:<port>`. Contact your ODT representative for the current host address.

SCHEDULE

FY27 Annual Training Calendar — SL 2 / SL 3 / FBC

Dual-team • 4-week cycle • 60 classes • 5 locations • ~450 personnel/year

⚑ SAMPLE SCHEDULE

This is a notional planning schedule for demonstration purposes. Dates, instructors, locations, and seat availability are not real. Actual class schedules will be published via official training channels.

MSS-TRMS — TRAINING RESOURCE MANAGEMENT SYSTEM

USAREUR-AF C2DAO

FY: **2027** | PROGRAM: **MSS-POI-001** | CYCLE: **4-WK** | MODE: **DUAL-TEAM** | CLASSES: **68** | STATUS: **PLANNING**

OIC — **1× Officer in Charge** — stationed Wiesbaden — scheduling, coord

ALPHA — **2× Instructors** — stationed Wiesbaden — travels to Poznań, Kaiserslautern, Vicenza, Ft. Hood MTT

BRAVO — **2× Instructors** — stationed Wiesbaden — travels to Graf, Ansbach, Vicenza

68
TOTAL CLASSES
32 SL 2 • 20 SL 3 • 8 FBC • 8 EXEC

1004
TOTAL SEATS
~490 unique personnel

7
LOCATIONS (MSCs)
4 DE • 1 IT • 1 PL • 1 CONUS

5
INSTRUCTORS
1 OIC • 2 A • 2 B

300
TRAINING DAYS
OCT 26 – JUL 27

60/40 split: SL 2 heavy — build the base first | **FBC 2×/quarter** | **EXEC 2×/quarter** (off weeks) | **7 sites (all MSCs):** Wiesbaden • Grafenwöhr • Vicenza • Ansbach • Kaiserslautern • Poznań • Ft. Hood MTT | SL 1 at unit level

CLASS NR	COURSE	TEAM	DATES	LOCATION	POC
001-FY27	SL 2	ALPHA	05–09 OCT 26	Wiesbaden, Clay Kaserne	SFC Chen
002-FY27	SL 2	BRAVO	05–09 OCT 26	Grafenwöhr, Bldg 244	SSG Williams
003-FY27	SL 2	ALPHA	13–17 OCT 26	Wiesbaden, Clay Kaserne	SFC Chen
004-FY27	SL 2	BRAVO	13–17 OCT 26	Grafenwöhr, Bldg 244	SSG Williams
005-FY27	SL 3	ALPHA	20–24 OCT 26	Wiesbaden, Clay Kaserne	CW3 Thomps
006-FY27	SL 3	BRAVO	20–24 OCT 26	Grafenwöhr, Bldg 244	CW2 Rodrigu
EX1-FY27	EXEC	OIC	28 OCT 26	Wiesbaden, Clay Kaserne	OIC
007-FY27	SL 2	ALPHA	02–06 NOV 26	Wiesbaden, Clay Kaserne	SFC Chen
008-FY27	SL 2	BRAVO	02–06 NOV 26	Grafenwöhr, Bldg 244	SSG Williams
009-FY27	SL 3	ALPHA	09–13 NOV 26	Wiesbaden, Clay Kaserne	CW3 Thomps
010-FY27	SL 3	BRAVO	09–13 NOV 26	Grafenwöhr, Bldg 244	CW2 Rodrigu
011-FY27	FBC #1	ALL	16–20 NOV 26	Wiesbaden, Clay Kaserne	OIC
EX2-FY27	EXEC	OIC	24 NOV 26	Wiesbaden, Clay Kaserne	OIC
012-FY27	SL 2	ALPHA	30 NOV–04 DEC 26	Vicenza, Caserma Del Din	SFC Chen

013-FY27	**SL 2**	BRAVO	30 NOV–04 DEC 26	Wiesbaden, Clay Kaserne	SSG Williams
014-FY27	**SL 3**	ALPHA	07–11 DEC 26	**Vicenza, Caserma Del Din**	CW3 Thomps
015-FY27	**FBC #2**	ALL	14–18 DEC 26	Wiesbaden, Clay Kaserne	OIC
016-FY27	**SL 2**	ALPHA	04–08 JAN 27	Wiesbaden, Clay Kaserne	SFC Chen
017-FY27	**SL 2**	BRAVO	04–08 JAN 27	Grafenwöhr, Bldg 244	SSG Williams
018-FY27	**SL 2**	ALPHA	11–15 JAN 27	Wiesbaden, Clay Kaserne	SFC Chen
019-FY27	**SL 2**	BRAVO	11–15 JAN 27	Grafenwöhr, Bldg 244	SSG Williams
020-FY27	**SL 3**	ALPHA	18–22 JAN 27	Wiesbaden, Clay Kaserne	CW3 Thomps
021-FY27	**SL 3**	BRAVO	18–22 JAN 27	Grafenwöhr, Bldg 244	CW2 Rodrigu
EX3-FY27	**EXEC**	OIC	27 JAN 27	Wiesbaden, Clay Kaserne	OIC
022-FY27	**SL 2**	ALPHA	01–05 FEB 27	Ansbach, Storck Bks	SFC Chen
023-FY27	**SL 2**	BRAVO	01–05 FEB 27	Grafenwöhr, Bldg 244	SSG Williams
024-FY27	**SL 3**	ALPHA	08–12 FEB 27	Ansbach, Storck Bks	CW3 Thomps
025-FY27	**SL 3**	BRAVO	08–12 FEB 27	Grafenwöhr, Bldg 244	CW2 Rodrigu
026-FY27	**FBC #3**	ALL	15–19 FEB 27	Wiesbaden, Clay Kaserne	OIC
EX4-FY27	**EXEC**	OIC	24 FEB 27	Wiesbaden, Clay Kaserne	OIC
027-FY27	**SL 2**	ALPHA	01–05 MAR 27	**Ft. Hood, TX (III Corps MTT)**	SFC Chen
028-FY27	**SL 2**	BRAVO	01–05 MAR 27	**Vicenza, Caserma Del Din**	SSG Williams
029-FY27	**SL 3**	ALPHA	08–12 MAR 27	**Ft. Hood, TX (III Corps MTT)**	CW3 Thomps
030-FY27	**FBC #4**	ALL	15–19 MAR 27	Wiesbaden, Clay Kaserne	OIC
031-FY27	**SL 2**	ALPHA	29 MAR–02 APR 27	Wiesbaden, Clay Kaserne	SFC Chen
032-FY27	**SL 2**	BRAVO	29 MAR–02 APR 27	Grafenwöhr, Bldg 244	SSG Williams
033-FY27	**SL 2**	ALPHA	05–09 APR 27	Wiesbaden, Clay Kaserne	SFC Chen
034-FY27	**SL 2**	BRAVO	05–09 APR 27	Grafenwöhr, Bldg 244	SSG Williams
035-FY27	**SL 3**	ALPHA	12–16 APR 27	Wiesbaden, Clay Kaserne	CW3 Thomps
036-FY27	**SL 3**	BRAVO	12–16 APR 27	Grafenwöhr, Bldg 244	CW2 Rodrigu
EX5-FY27	**EXEC**	OIC	21 APR 27	Wiesbaden, Clay Kaserne	OIC
037-FY27	**SL 2**	ALPHA	03–07 MAY 27	**Vicenza, Caserma Del Din**	SFC Chen
038-FY27	**SL 2**	BRAVO	03–07 MAY 27	Ansbach, Storck Bks	SSG Williams
039-FY27	**SL 3**	ALPHA	10–14 MAY 27	**Vicenza, Caserma Del Din**	CW3 Thomps
040-FY27	**SL 3**	BRAVO	10–14 MAY 27	Ansbach, Storck Bks	CW2 Rodrigu
041-FY27	**FBC #5**	ALL	17–21 MAY 27	Wiesbaden, Clay Kaserne	OIC
EX6-FY27	**EXEC**	OIC	26 MAY 27	Wiesbaden, Clay Kaserne	OIC
042-FY27	**SL 2**	ALPHA	31 MAY–04 JUN 27	Wiesbaden, Clay Kaserne	SFC Chen

042-FY27	SL 2	ALPHA	31 MAY–04 JUN 27	Wiesbaden, Clay Kaserne	SFC Chen
043-FY27	SL 2	BRAVO	31 MAY–04 JUN 27	Wiesbaden, Clay Kaserne	SSG Williams
044-FY27	SL 3	ALPHA	07–11 JUN 27	Wiesbaden, Clay Kaserne	CW3 Thomps
045-FY27	FBC #6	ALL	14–18 JUN 27	Wiesbaden, Clay Kaserne	OIC
046-FY27	SL 2	ALPHA	28 JUN–02 JUL 27	Wiesbaden, Clay Kaserne	SFC Chen

• NOT FINDING WHAT YOU NEED?

Contact your unit data steward for additional publications, source files, or access to restricted materials. For technical support, visit the **Support page →** For task-level procedures, use the **Task Index →**

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